An Improved Upper Bound for Colorings Without Symmetrically Colored k-Term Arithmetic Progressions

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Abstract

Given a coloring c and an even $k \geq 4$, a nontrivial k-term arithmetic progression (k-AP) $a, a+d, \ldots, a+(k-1)d$ is called *symmetrically colored* if $c(a+(i-1)d) = c(a+(k-i)d)$, $\forall i \in [k/2]$. Deng, Tidor, and Zhao asked whether $[N]$ admits a coloring with $N^{o(1)}$ colors and no such 4-APs, and gave an $O(N^{\log_{22} 3})$-coloring of $[N]$. We give an $O_k(p)$-coloring of $\mathbb{Z}/p^{k^2/4}\mathbb{Z}$ without such k-APs for every even $k \geq 4$ and every prime $p > k$, and hence an $O(N^{4/k^2})$-coloring of $[N]$, improving the exponent in the upper bound for 4-APs from $\log_{22} 3$ to $1/4$. The construction combines a carry-control coloring of base-p digits with a layered field norm mapping. Together with Behrend-style product colorings, our result for 4-APs gives $h(N) \leq N^{1/4+o(1)}$ in Erdős's Problem 160 on coloring every nontrivial 4-AP with at least three colors. This result also yields $\rho_4(\alpha) = O_\varepsilon(\alpha^{5-\varepsilon})$ for every $\varepsilon > 0$, improving the bound toward Ruzsa's question. Our result for k-APs disproves Gowers' conjectured lower bound for all even $k \geq 6$ for the *first* time.

1 Introduction

Given a coloring c and an even $k \geq 4$, we say that a k-term arithmetic progression (k-AP)

$$a, \quad a+d, \quad \ldots, \quad a+(k-1)d$$

is *nontrivial* if $d \neq 0$, and *symmetrically colored* if

$$c(a+(i-1)d) = c(a+(k-i)d) \qquad \forall i \in [k/2] = \{1, \ldots, k/2\} \, .$$

Deng, Tidor, and Zhao [DTZ23, Conjecture 1.6] proposed the following Ramsey-type problem: study colorings of $[N]$ without nontrivial symmetrically colored 4-APs, and conjectured that $N^{o(1)}$ colors suffice. Their tensor-power construction, based on a three-coloring of $\mathbb{Z}/22\mathbb{Z}$, yields

$$O\left(N^{\log_{22} 3}\right) \, ,$$

where $\log_{22} 3 \approx 0.355$. To the best of our knowledge, this is the strongest previously known upper bound for this problem. Our main result improves this to $O(N^{1/4})$ via an explicit construction.

Theorem 1.1. *For every even $k \geq 4$ and every prime $p > k$, there is a $(k-1)^{k^2/4-1}p$-coloring of $\mathbb{Z}/p^{k^2/4}\mathbb{Z}$ with no nontrivial symmetrically colored k-APs.*

Corollary 1.2. *For every prime $p \geq 5$, there is a $27p$-coloring of $\mathbb{Z}/p^4\mathbb{Z}$ with no nontrivial symmetrically colored 4-APs.*

This has several immediate applications. The first concerns Erdős Problem 160 [Erd89]:

> *Let $h(N)$ be the smallest k such that $[N]$ can be colored with k colors so that every 4-AP contains at least three distinct colors. Estimate $h(N)$.*

Hunter [Hun25] observed that the standard Behrend-style product coloring eliminates all relevant two-color patterns except the symmetric pattern. Combining this observation with our result improves the exponent in the previously known upper bound from $0.355 + o(1)$ to $1/4 + o(1)$.

Corollary 1.3 (An improved upper bound for Erdős Problem 160)**.** *One has*

$$h(N) \leq N^{1/4+o(1)} .$$

Second, let $\rho_4(\alpha)$ denote the minimum asymptotic density of 4-APs in a Fourier-uniform set of density α, in the sense of [DTZ23, Definition 1.3]. Ruzsa [CL07, Problem 3.2] asked whether $\rho_4(\alpha)$ decays faster than every power of α. Combining the reduction of [DTZ23] with our result improves the exponent in the previously known upper bound from 4.406 to arbitrarily close to 5.

Corollary 1.4 (An improved upper bound for Ruzsa's question)**.** *For every $\varepsilon > 0$, one has*

$$\rho_4(\alpha) = O_\varepsilon\big(\alpha^{5-\varepsilon}\big) \qquad (0 < \alpha < 1/2) .$$

For every even $k \geq 6$, this yields, to the best of our knowledge, the first counterexample to Gowers' conjectured lower bound; the case $k = 4$ was previously disproved by [Gow20, Theorem 6].

Corollary 1.5. *Gowers' conjectured lower bound [Gow01, Conjecture 4.2] is false for every even integer $k \geq 4$.*

2 From k-APs to digitwise k-APs

Fix an even $k \geq 4$, and a prime $p > k$. For $x \in \{0, 1, \dots, p-1\}$, define

$$\tau(x) := \left\lfloor \frac{(k-1)x}{p} \right\rfloor \in \{0, 1, \dots, k-2\} .$$

Thus τ divides $\{0, 1, \dots, p-1\}$ into $k-1$ consecutive intervals, each of diameter less than $p/(k-1)$.

Lemma 2.1. *Let $x_0, x_1, \dots, x_{k-1} \in \{0, 1, \dots, p-1\}$ satisfy*

$$x_{i+1} \equiv x_i + s \pmod{p} \qquad (i = 0, 1, \dots, k-2)$$

for some $s \in \{0, \dots, p-1\}$. For $i = 0, 1, \dots, k-2$, define the carry

$$e_i := \mathbf{1}_{\{x_{i+1} < x_i\}} .$$

If $\tau(x_{i-1}) = \tau(x_{k-i})$ for every $i \in [k/2]$, then $e_0 = e_1 = \ldots = e_{k-2}$.

Proof. If $s < (k-2)p/(k-1)$, then, since $\tau(x_{k/2-1}) = \tau(x_{k/2})$, necessarily $s < p/(k-1)$. If $s \geq (k-2)p/(k-1)$, we instead consider the reversed order $(x'_0, x'_1, \dots, x'_{k-1}) = (x_{k-1}, x_{k-2}, \dots, x_0)$ with $s' = p - s < p/(k-1)$, and

$$e'_i := \mathbf{1}_{\{x'_{i+1} < x'_i\}} = 1 - \mathbf{1}_{\{x_{k-1-i} < x_{k-2-i}\}} \qquad (i = 0, 1, \dots, k-2) .$$

Thus it suffices to consider the case $s < p/(k-1)$.

Let $\tau(x_{k/2-1}) = \tau(x_{k/2}) = i$. Since the step is less than $p/(k-1)$, the point $x_{k/2-2}$ lies either in interval i or $i-1$, while $x_{k/2+1}$ lies either in interval i or $i+1$ (interval indices understood modulo $k-1$). As $\tau(x_{k/2-2}) = \tau(x_{k/2+1})$, they must both lie in the same interval i. By simple induction we have that all of these points lie in the same interval i, and $e_0 = e_1 = \ldots = e_{k-2} = 0$. In the reversed case, the same argument gives $e'_i = 0$ for every i, and therefore $e_0 = e_1 = \cdots = e_{k-2} = 1$. □

Lemma 2.2 (digitwise k-AP)**.** *Let $L \geq 2$, and let*

$$n_i = a + id \in \mathbb{Z}/p^L\mathbb{Z} \qquad (i = 0, 1, \ldots, k-1) .$$

For each $n \in \mathbb{Z}/p^L\mathbb{Z}$, write its standard representative as

$$n = \sum_{j=0}^{L-1} x_j(n) p^j , \qquad x_j(n) \in \{0, 1, \ldots, p-1\} ,$$

and set

$$\mathbf{x}(n) := (x_0(n), \ldots, x_{L-1}(n)) \in \mathbb{F}_p^L .$$

Suppose that for every $j = 0, \ldots, L-2$,

$$\tau(x_j(n_{i-1})) = \tau(x_j(n_{k-i})) \qquad \forall i \in [k/2] .$$

Then there exists $\mathbf{h} \in \mathbb{F}_p^L$ such that

$$\mathbf{x}(n_i) = \mathbf{x}(n_0) + i\mathbf{h} \qquad (i = 0, 1, \ldots, k-1) .$$

Moreover, if $d \neq 0$ in $\mathbb{Z}/p^L\mathbb{Z}$, then $\mathbf{h} \neq 0$.

Proof. Iterating Lemma 2.1 from digit 0 to $L-2$ shows inductively that the $k-1$ carries to the next digit are equal at every digit position, and hence the digit vectors $\mathbf{x}(n_0), \ldots, \mathbf{x}(n_{k-1})$ form a k-AP in $\mathbb{F}_p^L$. And $\mathbf{h} = 0$ would imply $n_0 = n_1$ and thus $d = 0$. □

3 The field norm construction

Choose an irreducible polynomial $f(T) \in \mathbb{F}_p[T]$ of odd degree $t \geq 3$, and let α be one of its roots in $\mathbb{F}_{p^t}$. Then $\{1, \alpha, \ldots, \alpha^{t-1}\}$ is an $\mathbb{F}_p$-basis of $\mathbb{F}_{p^t}$. Identifying $(x_0, x_1, \ldots, x_{t-1}) \in \mathbb{F}_p^t$ with

$$x_0 + x_1\alpha + \ldots + x_{t-1}\alpha^{t-1} ,$$

define

$$P_t(x_0, x_1, \ldots, x_{t-1}) := \mathrm{N}_{\mathbb{F}_{p^t}/\mathbb{F}_p}\big(x_0 + x_1\alpha + \ldots + x_{t-1}\alpha^{t-1}\big) ,$$

where

$$\mathrm{N}_{\mathbb{F}_{p^t}/\mathbb{F}_p}(z) := z^{1+p+\ldots+p^{t-1}}.$$

The existence of $f(T)$, as well as the fact that $P_t : \mathbb{F}_p^t \to \mathbb{F}_p$ is a *homogeneous* polynomial of *degree* t with *no nontrivial zero*, is standard in the theory of finite fields; see [LN96, Corollary 2.11; Theorem 2.28]. We also let $P_1 : \mathbb{F}_p \to \mathbb{F}_p$ be the identity map.

4 The layered norm construction

For $\mathbf{x} = (x_0, x_1, \dots, x_{k^2/4-1}) \in \mathbb{F}_p^{k^2/4}$, we decompose

$$\mathbf{x} = \left(\mathbf{u}^{(1)}, \mathbf{u}^{(3)}, \dots, \mathbf{u}^{(k-1)}\right) ,$$

where $\mathbf{u}^{(i)} := \left(x_{(i-1)^2/4}, x_{(i-1)^2/4+1}, \dots, x_{(i+1)^2/4-1}\right) \in \mathbb{F}_p^i$. Then we define

$$Q(\mathbf{x}) := \sum_{1\le i\le k-1,\ i \text{ odd}} P_i(\mathbf{u}^{(i)}) .$$

Lemma 4.1 (layered norm)**.** *Let*

$$\mathbf{y}_i = \mathbf{y}_0 + i\mathbf{h} \qquad (i = 0, 1, \dots, k-1)$$

be a k-AP in $\mathbb{F}_p^{k^2/4}$. If $Q(\mathbf{y}_{i-1}) = Q(\mathbf{y}_{k-i})$ for every $i \in [k/2]$, then $\mathbf{h} = 0$.

Proof. Decompose

$$\mathbf{y}_0 = \left(\mathbf{u}^{(1)}, \mathbf{u}^{(3)}, \dots, \mathbf{u}^{(k-1)}\right) , \qquad \mathbf{h} = \left(\mathbf{v}^{(1)}, \mathbf{v}^{(3)}, \dots, \mathbf{v}^{(k-1)}\right) ,$$

where $\mathbf{u}^{(i)}, \mathbf{v}^{(i)} \in \mathbb{F}_p^i$ for every odd $i \in \{1, 3, \dots, k-1\}$.

Let $a := \frac{k-1}{2} \in \mathbb{F}_p$, where division by 2 is taken in $\mathbb{F}_p$, and define

$$g(t) := Q(\mathbf{y}_0 + t\mathbf{h}) = \sum_{1\le i\le k-1,\ i \text{ odd}} P_i\big(\mathbf{u}^{(i)} + t\mathbf{v}^{(i)}\big) \in \mathbb{F}_p[t] .$$

Recall that $Q(\mathbf{y}_{i-1}) = Q(\mathbf{y}_{k-i})$ for every $i \in [k/2]$, so $g(t+a) - g(-t+a)$ has at least k roots in $\mathbb{F}_p$: $\{-a, -a+1, \dots, k-1-a\}$, which are distinct because $p > k$. Note that $\deg g$ is at most $k-1$, and by the standard fact that a nonzero polynomial of degree at most $k-1$ over a field has at most $k-1$ roots, $g(t+a) - g(-t+a)$ must be the zero polynomial, *i.e.*, $g(t+a)$ is an even polynomial in t.

We now prove by descending induction over the odd integers $r = k-1, k-3, \dots, 1$ that $\mathbf{v}^{(r)} = 0$. Suppose that $\mathbf{v}^{(s)} = 0$ for every odd $s > r$. The corresponding terms

$$P_s\big(\mathbf{u}^{(s)} + (t+a)\mathbf{v}^{(s)}\big) = P_s\big(\mathbf{u}^{(s)}\big)$$

are then constant in t, while every term indexed by $s < r$ has degree less than r. Then by the homogeneity of P_r, the coefficient of t^r in $g(t+a)$ is exactly $P_r\big(\mathbf{v}^{(r)}\big)$. Since $g(t+a)$ is an even polynomial in t and r is odd, this coefficient must vanish. Therefore $P_r\big(\mathbf{v}^{(r)}\big) = 0$. As P_r has no nontrivial zero, it follows that $\mathbf{v}^{(r)} = 0$. The induction therefore gives

$$\mathbf{h} = \big(\mathbf{v}^{(1)}, \mathbf{v}^{(3)}, \dots, \mathbf{v}^{(k-1)}\big) = 0 .$$

□

Proof of Theorem 1.1. For $n \in \mathbb{Z}/p^{k^2/4}\mathbb{Z}$, write its standard base-p expansion as

$$n = x_0(n) + px_1(n) + \dots + p^{k^2/4-1}x_{k^2/4-1}(n) .$$

Let $\mathbf{x}(n) := (x_0(n), x_1(n), \dots, x_{k^2/4-1}(n))$. Define

$$c_k(n) := \big(\tau(x_0(n)), \dots, \tau(x_{k^2/4-2}(n)), Q(\mathbf{x}(n))\big) .$$

This coloring uses at most $(k-1)^{k^2/4-1}p$ colors. Suppose that

$$n_i = a + id\,, \qquad i = 0, 1, \ldots, k-1\,,$$

is symmetrically colored by c_k. Applying Lemma 2.2 with $L = k^2/4$, we obtain

$$\mathbf{x}(n_i) = \mathbf{x}(n_0) + i\mathbf{h}$$

for some $\mathbf{h} \in \mathbb{F}_p^{k^2/4}$. Lemma 4.1 then implies that $\mathbf{h} = 0$. Lemma 2.2 therefore gives $d = 0$ in $\mathbb{Z}/p^{k^2/4}\mathbb{Z}$. Hence there is no nontrivial symmetrically colored k-AP. $\square$

Corollary 4.2. *For every fixed even integer $k \geq 4$ and every positive integer N, the interval $[N]$ admits a coloring with $O_k(N^{4/k^2})$ colors and no nontrivial symmetrically colored k-AP. More precisely, for all sufficiently large N, at most $2(k-1)^{k^2/4-1}N^{4/k^2}$ colors suffice.*

Proof. By the Bertrand–Chebyshev theorem, for every integer $N \geq k^{k^2/4}$, there is a prime $p > k$ with

$$\lfloor N^{4/k^2} \rfloor < p < 2\lfloor N^{4/k^2} \rfloor\,.$$

Restricting the coloring from Theorem 1.1 to $[N] \subseteq \mathbb{Z}/p^{k^2/4}\mathbb{Z}$ gives the result. $\square$

5 Application to Erdős Problem 160

In Erdős Problem 160, one must exclude not only the symmetric $ABBA$ pattern, but all color patterns on a nontrivial 4-AP that use at most two colors. Hunter [Hun25] observed that the remaining patterns can be eliminated by taking the product with Behrend-style colorings [Beh46] using $N^{o(1)}$ colors. We give a proof of this observation.

Lemma 5.1 (Lemma 3.2 of [DTZ23])**.** *The patterns $AAAA$, $AAAB$, $AABA$, $ABAA$, and $BAAA$, on nontrivial 4-APs in $[N]$, can be eliminated using an $N^{o(1)}$ coloring.*

Lemma 5.2 (Lemma 7.11 of [DTZ23])**.** *The pattern $ABAB$ on nontrivial 4-APs in $[N]$ can be eliminated using an $N^{o(1)}$ coloring.*

Lemma 5.3 (Behrend-style coloring)**.** *The pattern $AABB$ on nontrivial 4-APs in $[N]$ can be eliminated using an $N^{o(1)}$ coloring.*

Proof. Let $M = 2^{\lceil\sqrt{\log_2 N}\rceil}$ and $m = \lceil \log_M(N+1) \rceil$. For $n \in [N]$, write its standard base-M expansion as $n = \sum_{j=0}^{m-1} x_j(n)M^j$. Define $\tau(x) := \lfloor \log_2(M-x) \rfloor$, $\Psi(n) := \sum_{j=0}^{m-1} x_j(n)^2$, and color n by $\psi_N(n) := \big(\tau(x_0(n)), \ldots, \tau(x_{m-1}(n)), \Psi(n)\big)$. The total number is $O((\log M)^m mM^2) = N^{o(1)}$.

We first prove that the carries are consistent. Suppose y_0, y_1, y_2, y_3 form a 4-AP in $\mathbb{Z}/M\mathbb{Z}$ and $\tau(y_0) = \tau(y_1)$, $\tau(y_2) = \tau(y_3)$. If the step size $s > M/2$, then we can reverse the order as in the proof of Lemma 2.1. So we only need to focus on $s \leq M/2$. A carry in the first or third addition would place the corresponding pair on opposite sides of $M/2$, so $\tau(y_0) \neq \tau(y_1)$ or $\tau(y_2) \neq \tau(y_3)$. If only the middle addition carried, then $s = y_1 - y_0 \geq M - y_1$, so $M - y_0 \geq 2(M - y_1)$, and thus $\tau(y_0) \geq \tau(y_1) + 1$, leading to contradiction.

Now let n_0, n_1, n_2, n_3 be a 4-AP with color pattern $AABB$. The carry consistency established above yields a digitwise 4-AP $\mathbf{x}(n_i) = \mathbf{u} + i\mathbf{v}$, where $i \in \{0, 1, 2, 3\}$, for some $\mathbf{u}, \mathbf{v} \in \mathbb{Z}^m$. Let $f(t) := \|\mathbf{u}\|^2 + 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2\|\mathbf{v}\|^2$, which is a quadratic function. However, the AABB pattern indicates that f has two axes of symmetry, $1/2$ and $5/2$, therefore f must be a constant function, and thus $\mathbf{v} = 0$. Consequently, every 4-AP with color pattern $AABB$ is trivial. $\square$

Proof of Corollary 1.3. Taking the product of the coloring from Corollary 4.2 with $k = 4$ and the colorings in Lemmas 5.1, 5.2, and 5.3 uses $N^{1/4+o(1)}$ colors. $\square$

6 Consequence for Fourier-uniform sets

Deng, Tidor, and Zhao [DTZ23] proved that an r-coloring of $\mathbb{Z}/M\mathbb{Z}$ without symmetrically colored 4-APs implies

$$\rho_4(\alpha) = O_{r,M}(\alpha^{3+\frac{1}{2}\log_r M}) \,, \tag{1}$$

for $0 < \alpha < 1/2$; see [DTZ23, Theorem 1.8]. They also proved that an r-coloring of $\mathbb{Z}/M\mathbb{Z}$ without symmetrically colored k-APs implies

$$\rho_k(\alpha) = O_{r,M,k,\varepsilon}\left(\alpha^{k-1+(\log_r M)/(k-1)-\varepsilon}\right) \,, \tag{2}$$

for even $k \geq 4$ and $0 < \alpha < 1$; see [DTZ23, Proposition 2.4 and Theorem 2.8].

Proof of Corollary 1.4. By Theorem 1.1, the exponent in (1) can be taken to be

$$3 + \frac{1}{2}\log_{27p}(p^4) = 3 + \frac{2\log p}{\log p + \log 27} \,.$$

This tends to 5 as $p \to \infty$. □

Proof of Corollary 1.5. Applying (2) with $M = p^{k^2/4}$ and $r = (k-1)^{k^2/4-1}p$ gives that for every even $k \geq 4$ and every $\varepsilon > 0$, the exponent can be taken to be

$$k - 1 + \frac{\frac{k^2}{4}\log p}{(k-1)\left(\left(\frac{k^2}{4}-1\right)\log(k-1) + \log p\right)} - \varepsilon \,.$$

As $p \to \infty$, this tends to

$$k + \frac{(k-2)^2}{4(k-1)} - \varepsilon \,.$$

Choosing $\varepsilon < \frac{(k-2)^2}{4(k-1)}$ makes the exponent larger than the random exponent k. This provides the first counterexamples to Gowers' conjectured lower bound [Gow01, Conjecture 4.2] for every even $k \geq 6$. □

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